

defines who may communicate and who may not. Yet Congress and the FCC routinely determine who may broadcast on certain frequencies, and they regularly shut down those, such as “pirate” radio broadcasters, who fail to observe those rules.

The rationale for limiting speech over the airwaves is that there is no alternative. Spectrum is scarce, so the argument goes, so either some may speak or none will be able to get their message across amid the cacophony of interfering voices. As discussed above, though, that scarcity is not an immutable property of a physical spectrum resource. It’s a historically and technologically contingent judgment. Needless to say, much has changed since the 1920s. And indeed, there has been a major shift in the government’s approach to spectrum assignment. Auctions have replaced outright grants, competitive hearings, and lotteries as the tool of choice. Beginning with Ronald Coase’s seminal 1959 article, “The Federal Communications Commission,” economists have argued persuasively that competitive bidding is the most efficient way to assign scarce licenses among competing users. Starting with the personal communications service (PCS) auctions of 1994, the FCC has raised over \$30 billion for the US Treasury and delivered substantial new spectrum to the marketplace in this manner.

Most of the debates around spectrum policy today involve variations of the auction idea. Some parties advocate secondary markets or moving from licenses to fee simple ownership, while others propose combining auctions with annual lease fees after the initial license period. These debates, intense though they may be, occur within the safe confines of the dominant exclusive licensing paradigm. If we have decided to license wireless frequencies, there are important questions about how best to do so. But why take licensing for granted?

A Spectrum Commons

Capacity-magnifying techniques such as spread spectrum, cooperative networking and software-defined radio make it possible